# Abubakr M. Abdelhalim

Electrical engineer — hardware design, embedded systems, power and signal integrity

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Final-year electrical engineering student at Universiti Teknologi Malaysia, graduating February 2027. At Pythonium.ai I worked on the hardware design of a 4-channel industrial DAQ module, from part selection and simulation to stackup and schematic review. Gold Medal, MIIE 2026; Industry Award, UTM EE Capstone Showcase 2026.

## EXPERIENCE

**Embedded / PCB Design Engineer Intern** · Pythonium.ai — Jul – Oct 2026
Johor Bahru, Malaysia

- 4-channel industrial DAQ module: STM32H563 MCU, AD4111 ADC, AD3531 output DAC, LT3514 triple-buck power tree, LAN8742A Ethernet PHY, RS-485, MAX31865 RTD and 4–20 mA inputs.
- Replaced the ADE7913 with the AD4111 and the AD5422 with the AD3531; removed the digital isolator from the isolation scheme; stabilised the output stage with an isolation resistor and feedback capacitor, verified in simulation.
- Chose TU-872 over FR-4 and Rogers for the stackup; ran schematic reviews with findings logged by severity.
- Designed a passive M.2-to-EDSFF adapter linking an SSD tester to an on-board diagnostic card; built the footprints from connector spec dimensions and converted the board to rigid-flex.

**Part-Time Engineer** · TSH Group — Apr – Jul 2025
Factory automation and commissioning, Johor

- PLC testing and industrial wiring for automation systems; wrote the work instructions for MTC machine testing.
- Designed and built a DI/DO testing kit for verifying industrial PLC signals in the field.

**Electronics Head of Department** · UTM Robocon — Jan 2024 – Apr 2025
Malaysia's team at ABU Robocon 2024 and 2025

- Designed 4-layer STM32F4 main-control and STM32G4 motor-interface boards in Altium, with CAN, $I^2C$ and SPI links across all subsystems.
- Upgraded the mainboard's short-circuit protection and layout; designed a custom IMU board for navigation and STM32 + PID shooter control.
- Sole electronics lead between the mechanical and software teams; trained 20+ members in Altium and STM32.

## EDUCATION

**Bachelor of Electrical Engineering (Hons)** · Universiti Teknologi Malaysia — Mar 2023 – Feb 2027
Final-year project: supraharmonic emissions in PV inverter systems · Student member, Saudi Council of Engineers

## PROJECTS

**Smart Posture Correction System** · Capstone
GOLD MEDAL, MIIE 2026 · INDUSTRY AWARD, UTM EECS 2026

- Python/OpenCV and dlib face-pose pipeline fused with IMU data to classify posture; haptic feedback in under 200 ms.
- 4-layer KiCad PCB (XIAO ESP32-C3, MPU6050), C++ firmware over UDP with Blynk IoT, SolidWorks enclosure.

**Datasheet lookup tool** — 2026

- Local Python CLI for component datasheets: PyMuPDF extraction, SQLite FTS5 search, incremental indexing, IC tagging.

**Other builds**

- Self-balancing robot (ESP32, MPU6050, PID) · STM32 line follower with Bluetooth override · custom boards: isolated valve drivers, DC-DC bucks, ESP32 and STM32 dev boards.

## LEADERSHIP

- Organising committee and referee, IYRC 2025 (International Youth Robot Competition), Johor · Jun 2025
- Presenter and mentor, month-long Self-Balancing Robot Workshop: PCB design, embedded systems, PID · May 2025
- Electronics mentor, FIRST Global Challenge, Team Sudan

## SKILLS

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|---|---|
| HARDWARE DESIGN | Altium Designer, KiCad, 4-layer and rigid-flex PCB, stackup selection, schematic review, SolidWorks |
| SIMULATION | LTspice, PSpice, ngspice, MATLAB/Simulink, NI Multisim |
| FIRMWARE | Embedded C on STM32 (CubeMX, CubeIDE), peripheral bring-up and drivers, ESP32, CAN, $I^2C$, SPI, RS-485 |
| SOFTWARE | Python, C/C++, OpenCV, SQLite |
| INDUSTRIAL | PLC testing, industrial wiring, DI/DO signal verification |